

Demonstration of Control and User Plane Separation (CUPS) in the 5G Core Using PFCP

1. Aim

To demonstrate the Control and User Plane Separation (CUPS) architecture of the 5G Core network by capturing and analyzing the PFCP Session Establishment procedure exchanged between the Session Management Function (SMF) and the User Plane Function (UPF) over the N4 interface, using an Open5GS-based 5G Core deployment and Wireshark.

2. Objectives

1. To understand the architecture of the 5G Core network and the concept of Control and User Plane Separation.
2. To set up an Open5GS-based 5G Core network and a UERANSIM-based UE/gNB simulator.
3. To configure a subscriber and establish registration and a PDU session between the UE and the network.
4. To capture PFCP signaling messages exchanged between the SMF and the UPF using Wireshark.
5. To identify and interpret the PFCP Session Establishment Request and Response messages on the N4 interface.
6. To verify that the control plane and user plane functions of the 5G Core operate independently, in line with the CUPS principle.

3. Introduction

The 5G Core network follows a service-based architecture in which each network function performs a well-defined role through standardized interfaces. Control and User Plane Separation (CUPS) is a key architectural change that decouples signaling functions from user-data forwarding so each can be deployed, scaled, and upgraded independently. This separation is represented by the SMF on the control plane and the UPF on the user plane, allowing UPFs to be placed near the network edge to reduce latency while the SMF continues to manage sessions centrally. Since the SMF and UPF are separate functions, they coordinate through the Packet Forwarding Control Protocol (PFCP) over the N4 interface, through which the SMF installs rules in the UPF that define how user packets are detected, forwarded, and treated.

Wireshark, an open-source packet analyzer with built-in support for dissecting PFCP messages, provides an effective way to observe this relatively new protocol directly at the signaling level. This experiment builds a small-scale 5G Core using Open5GS, generates a real UE registration and PDU session request using the UERANSIM simulator, and uses Wireshark to capture the PFCP messages exchanged between the SMF and the UPF over N4 — verifying the operation of CUPS in practice.

4. Theory

4.1 5G Core Network Architecture

The 5G Core follows a Service-Based Architecture (SBA) in which network functions expose services through standardized, mostly HTTP-based interfaces, except for a few interfaces such as N4 that use a dedicated protocol. Key functions include the AMF (registration and mobility), SMF (PDU session management), UPF (user-data forwarding), AUSF (authentication), and UDM (subscriber data storage) — each deployable, scalable, and upgradable independently.

4.2 Control and User Plane Separation (CUPS)

The control plane handles signaling — registration, authentication, mobility, and session establishment — mainly via the AMF and SMF, while the user plane, represented by the UPF, handles only the transport of user data. CUPS is the principle of splitting these two roles into independent network functions; here it is represented by the SMF (session control) and UPF (packet forwarding), coordinated over the standardized N4 interface using PFCP. This separation enables independent scaling of user-plane capacity, flexible edge placement of UPFs, simpler upgrades to either plane, and better support for latency-sensitive use cases such as industrial automation or cloud gaming.

4.3 Session Management Function (SMF) and User Plane Function (UPF)

The SMF is the control-plane function responsible for the full lifecycle of a PDU session: it selects the appropriate UPF, allocates the UE's IP address, applies QoS policy, and generates the forwarding rules to be installed in the UPF. In this experiment, the SMF initiates the PFCP Association Setup and Session Establishment procedures. The UPF performs the actual forwarding of user packets strictly according to the Packet Detection Rules (PDRs), Forwarding Action Rules (FARs), and QoS Enforcement Rules (QERs) installed by the SMF, without making independent traffic decisions, and responds to the SMF's PFCP Session Establishment Request with a corresponding Response.

4.4 Packet Forwarding Control Protocol (PFCP) and the N4 Interface

PFCP is the protocol standardized by 3GPP for communication between the control plane and the user plane in the Evolved Packet Core and the 5G Core. It runs over UDP port 8805 and supports procedures including Association Setup, Session Establishment, Session Modification, Session Deletion, and Heartbeat, using Information Elements such as PDRs, FARs, and QERs to describe the rules to be applied. The N4 interface is the reference point connecting the SMF and the UPF and is used exclusively to carry this PFCP signaling — no user data is transported over N4. In this experiment, since the SMF and UPF were deployed on the same Ubuntu host, N4 was realized over the loopback interface, with the SMF and UPF bound to distinct loopback addresses communicating over UDP port 8805.

4.5 PDU Session Establishment (Overview)

A UE first registers with the network through the AMF and then requests a PDU session via the SMF. Once the SMF selects a UPF for the session, it configures the UPF through PFCP so the forwarding path is ready before any user data is sent; only after this is confirmed does the UE receive an IP address and begin sending data through the assigned UPF. This sequence illustrates why PFCP and the N4 interface are essential to session establishment.

5. Software Requirements

Software / Tool	Purpose in the Experiment
Ubuntu 22.04 LTS	Host operating system used to run the 5G Core network functions and the simulator.
VirtualBox	Virtualization platform used to host the Ubuntu environment.
Open5GS	Open-source 5G Core implementation providing the AMF, SMF, UPF, and other core network functions.
UERANSIM	Open-source 5G UE and gNB simulator used to generate registration and PDU session traffic.
Wireshark	Packet analyzer used to capture and inspect PFCP signaling over the N4 interface.

6. System Architecture

The experimental setup consists of a simulated UE and gNB (UERANSIM) and a set of 5G Core network functions (Open5GS), all running on a single Ubuntu host. The UE and gNB communicate over the NG-RAN protocol stack, and the gNB connects to the AMF via the N2 interface. The AMF, SMF, and other core functions communicate internally, while the SMF communicates with the UPF over the N4 interface using PFCP. Once a PDU session is established, user data flows from the UE through the gNB and the UPF towards the Internet, while control signaling continues to flow separately between the UE, the AMF, and the SMF.

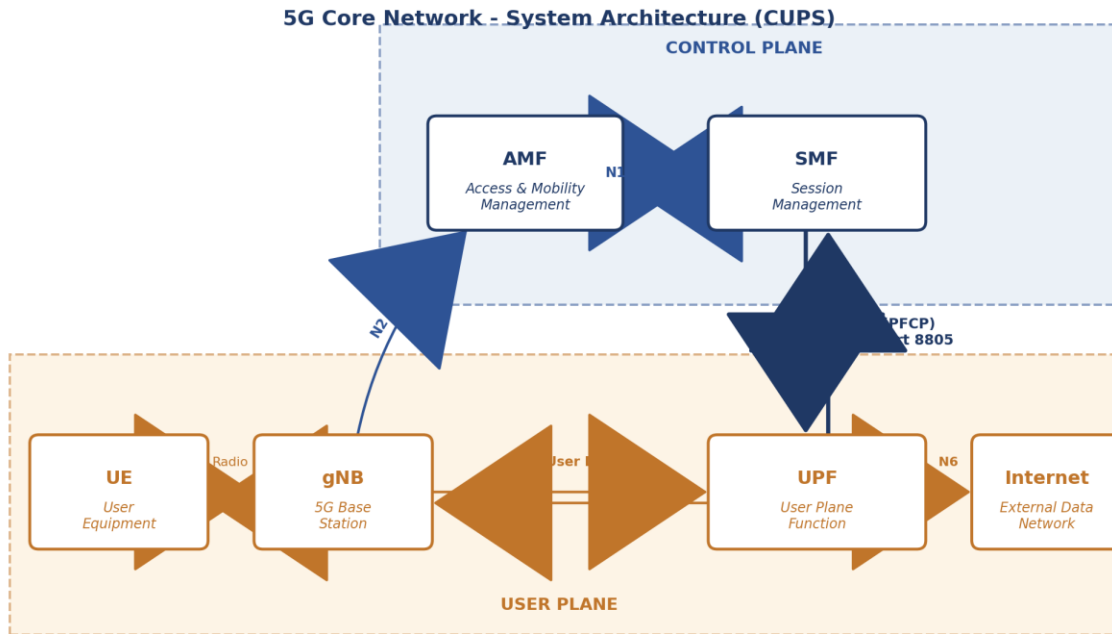


Figure 1 - 5G System Architecture showing UE, gNB, AMF, SMF, UPF, N4 Interface and Internet connectivity

The diagram separates the control plane (AMF, SMF) from the user plane (gNB, UPF, external data network); the SMF and UPF communicate exclusively over the N4 interface using PFCP over UDP port 8805 — the specific link examined in this experiment. The UE, simulated via UERANSIM's nr-ue application, initiates registration and requests a PDU session; the gNB, simulated via nr-gnb, provides radio connectivity and forwards NAS signaling to the AMF over N2 using SCTP. The AMF handles registration, authentication coordination, and mobility, forwarding session requests to the SMF, which manages the PDU session lifecycle and configures the UPF via PFCP (Section 4.3). The UPF forwards user-plane traffic per the rules installed by the SMF, and connects the UE's data session to the Internet once configuration is complete.

7. Procedure

7. Install Ubuntu 22.04 LTS on the working machine (or within VirtualBox).
8. Install and build Open5GS along with its required dependencies and MongoDB database.
9. Start the MongoDB service and confirm that it is running.
10. Start all Open5GS network function services, including the AMF, SMF, and UPF.
11. Open the Open5GS WebUI and configure a subscriber with the required IMSI, keys, and DNN settings.

12. Install and configure UERANSIM, then start the simulated gNB and observe its connection to the AMF.
13. Start the simulated UE using UERANSIM and observe the registration and PDU session establishment process.
14. Open Wireshark and select the loopback interface, since the SMF and UPF communicate over loopback addresses in this setup.
15. Apply a display filter for pfcf to isolate PFCP traffic from other captured packets.
16. Observe the PFCP Association Setup and Session Establishment Request/Response messages exchanged between the SMF and the UPF.
17. Save the packet capture file for documentation and further analysis.

8. Implementation

This section presents the actual outputs obtained while performing the experiment and explains what each output demonstrates.

Step 1 - Configure Subscriber

A subscriber profile must exist in the Open5GS database before any registration or session activity can take place. Using the Open5GS WebUI, a subscriber was added with the IMSI 999700000000001, its associated security keys (K and OPc), and an allocated DNN of "internet" under S-NSSAI SST 1. The WebUI confirmed the subscriber status as SERVICE_GRANTED, indicating the profile was active — a prerequisite for the entire experiment, since the AMF and SMF reject any UE not present in the subscriber database.

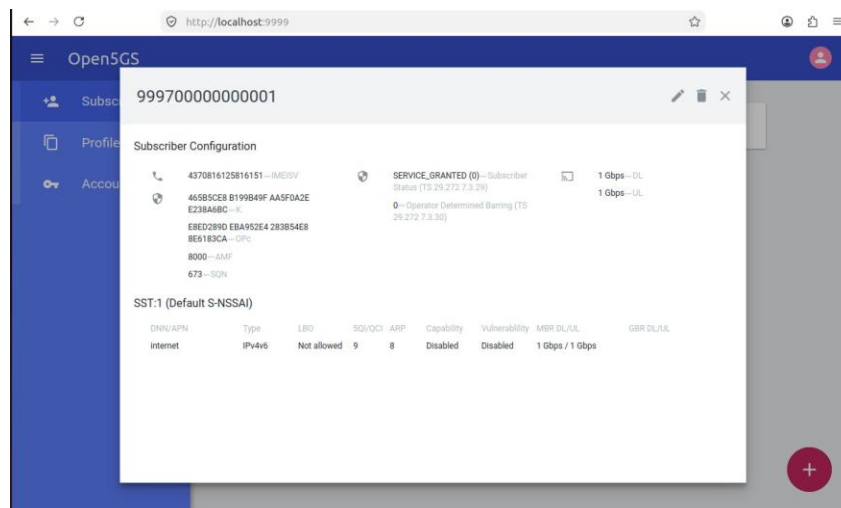


Figure 2 - Open5GS Subscriber Configuration

Step 2 - Start UERANSIM gNB

The simulated gNB was started next. Its terminal log shows an SCTP connection established to the AMF at 127.0.0.5 on port 38412 (the standard N2 interface port), followed by a successful NG Setup Request/Response exchange. Once a UE later attached, the gNB completed an RRC Setup for the UE, forwarded the initial NAS message to the AMF, and set up the PDU session resource after receiving the Initial Context Setup Request — confirming that the gNB correctly relayed signaling between the UE and the AMF.

```

UERANSIM v3.3.0
[2026-07-22 20:30:13.369] [sctp] [info] Trying to establish SCTP connection... (127.0.0.5:38412)
[2026-07-22 20:30:13.371] [sctp] [info] SCTP connection established (127.0.0.5:38412)
[2026-07-22 20:30:13.372] [sctp] [debug] SCTP association setup ascId[11]
[2026-07-22 20:30:13.373] [ngap] [debug] Sending NG Setup Request
[2026-07-22 20:30:13.431] [ngap] [debug] NG Setup Response received
[2026-07-22 20:30:13.431] [ngap] [info] NG Setup procedure is successful
[2026-07-22 20:30:37.701] [rrc] [debug] UE[1] new signal detected
[2026-07-22 20:30:37.726] [rrc] [info] RRC Setup for UE[1]
[2026-07-22 20:30:37.782] [ngap] [debug] Initial NAS message received from UE[1]
[2026-07-22 20:30:37.889] [ngap] [debug] Initial Context Setup Request received
[2026-07-22 20:30:38.206] [ngap] [info] PDU session resource(s) setup for UE[1] count[1]
[2026-07-22 20:32:32.530] [sctp] [debug] SCTP association shutdown (clientId: 2)
[2026-07-22 20:32:32.533] [sctp] [warning] Unhandled SCTP notification received
[2026-07-22 20:32:32.533] [ngap] [error] Association terminated for AMF[2]
[2026-07-22 20:32:32.533] [ngap] [debug] Removing AMF context[2]

```

Figure 3 - gNB Terminal Output

Step 3 - Register UE and Establish PDU Session

The UE log shows the complete registration and session sequence: an Initial Registration request, an RRC Setup Request/Response exchange bringing the UE into the RRC-CONNECTED state, an Authentication Request/Response, and an accepted Security Mode Command. The UE then received a Registration Accept, moved into the MM-REGISTERED/NORMAL-SERVICE state, and returned a Registration Complete message, marking Initial Registration as successful.

The UE subsequently sent a PDU Session Establishment Request and received a PDU Session Establishment Accept for PSI[1], confirming that the PDU session was established successfully. A TUN interface (uesimtun0) was brought up with IP address 10.45.0.3, representing the UE's user-plane data path — usable only once the underlying UPF has been configured, which is the process captured in the next steps.

```

[2026-07-22 20:36:17.811] [nas] [debug] UAC access attempt is allowed for identity[0], category[MO_sig]
[2026-07-22 20:36:17.811] [nas] [debug] Sending Initial Registration
[2026-07-22 20:36:17.812] [rrc] [debug] Sending RRC Setup Request
[2026-07-22 20:36:17.813] [nas] [info] UE switches to state [MM-REGISTER-INITIATED]
[2026-07-22 20:36:17.815] [rrc] [info] RRC connection established
[2026-07-22 20:36:17.815] [rrc] [info] UE switches to state [RRC-CONNECTED]
[2026-07-22 20:36:17.816] [nas] [info] UE switches to state [CM-CONNECTED]
[2026-07-22 20:36:17.875] [nas] [debug] Authentication Request received
[2026-07-22 20:36:17.875] [nas] [debug] Received SQN [000000000281]
[2026-07-22 20:36:17.875] [nas] [debug] SQN-MS [000000000000]
[2026-07-22 20:36:17.885] [nas] [debug] Security Mode Command received
[2026-07-22 20:36:17.885] [nas] [debug] Selected integrity[2] ciphering[0]
[2026-07-22 20:36:17.921] [nas] [debug] Registration accept received
[2026-07-22 20:36:17.921] [nas] [info] UE switches to state [MM-REGISTERED/NORMAL-SERVICE]
[2026-07-22 20:36:17.921] [nas] [debug] Sending Registration Complete
[2026-07-22 20:36:17.921] [nas] [info] Initial Registration is successful
[2026-07-22 20:36:17.921] [nas] [debug] Sending PDU Session Establishment Request
[2026-07-22 20:36:17.923] [nas] [debug] UAC access attempt is allowed for identity[0], category[MO_sig]
[2026-07-22 20:36:18.131] [nas] [debug] Configuration Update Command received
[2026-07-22 20:36:18.176] [nas] [debug] PDU Session Establishment Accept received
[2026-07-22 20:36:18.176] [nas] [info] PDU Session establishment is successful PSI[1]
[2026-07-22 20:36:18.359] [app] [info] Connection setup for PDU session[1] is successful, TUN interface[uesimtun0, 10.45.0.3] is up.

```

Figure 4 - UE Registration Output

Step 4 - Start Wireshark Capture

Wireshark was launched with root privileges and used to start and stop multiple capture sessions on an interface while the Open5GS core network functions were active. Each capture session was automatically written to a temporary pcapng file, as shown by the repeated Capture Start, Capture started, and File messages in the terminal output. This iterative process aligned the capture window with the timing of UE registration and the PDU session request, ensuring that the relevant PFCP exchange between the SMF and the UPF would be recorded.

```

abinaya@abinaya-VirtualBox: ~$ sudo wireshark
[sudo] password for abinaya:
** (wireshark:2864) 20:28:48.883802 [GUI WARNING] -- QStandardPaths: XDG_RUNTIME_DIR not set, defaulting to '/tmp/runtime-root'
** (wireshark:2864) 20:28:55.880111 [Capture MESSAGE] -- Capture Start ...
** (wireshark:2864) 20:28:55.952856 [Capture MESSAGE] -- Capture started
** (wireshark:2864) 20:28:55.952951 [Capture MESSAGE] -- File: "/tmp/wireshark_loUYECT3.pcapng"
** (wireshark:2864) 20:28:59.473066 [GUI WARNING] -- failed to create compose table
** (wireshark:2864) 20:30:58.752872 [Capture MESSAGE] -- Capture Stop ...
** (wireshark:2864) 20:30:58.802627 [Capture MESSAGE] -- Capture stopped.
** (wireshark:2864) 20:30:59.264522 [Capture MESSAGE] -- Capture Start ...
** (wireshark:2864) 20:30:59.445963 [Capture MESSAGE] -- Capture started
** (wireshark:2864) 20:30:59.447013 [Capture MESSAGE] -- File: "/tmp/wireshark_loAUC0S3.pcapng"
** (wireshark:2864) 20:31:48.887040 [Capture MESSAGE] -- Capture Stop ...
** (wireshark:2864) 20:31:48.904385 [Capture MESSAGE] -- Capture stopped.
** (wireshark:2864) 20:31:55.292456 [Capture MESSAGE] -- Capture Start ...
** (wireshark:2864) 20:31:55.417515 [Capture MESSAGE] -- Capture started
** (wireshark:2864) 20:31:55.417584 [Capture MESSAGE] -- File: "/tmp/wireshark_loS891S3.pcapng"
** (wireshark:2864) 20:34:13.786963 [Capture MESSAGE] -- Capture Stop ...
** (wireshark:2864) 20:34:13.816372 [Capture MESSAGE] -- Capture stopped.
** (wireshark:2864) 20:34:13.897782 [Capture MESSAGE] -- Capture Start ...
** (wireshark:2864) 20:34:14.009864 [Capture MESSAGE] -- Capture started
** (wireshark:2864) 20:34:14.010161 [Capture MESSAGE] -- File: "/tmp/wireshark_loFRHVS3.pcapng"
** (wireshark:2864) 20:35:07.042667 [Capture MESSAGE] -- Capture Stop ...
** (wireshark:2864) 20:35:07.084366 [Capture MESSAGE] -- Capture stopped.
** (wireshark:2864) 20:35:07.200349 [Capture MESSAGE] -- Capture Start ...
** (wireshark:2864) 20:35:07.316707 [Capture MESSAGE] -- Capture started
** (wireshark:2864) 20:35:07.316768 [Capture MESSAGE] -- File: "/tmp/wireshark_lo1QZYS3.pcapng"

```

Figure 5 - Wireshark Capture Started

Step 5 - Capture PFCP Session Establishment

With the pfcf display filter applied and the loopback interface selected, Wireshark clearly isolated the PFCF exchange between the SMF and the UPF: a PFCF Session Establishment Request sent from 127.0.0.4 (SMF) to 127.0.0.7 (UPF), immediately followed by a PFCF Session Establishment Response sent back from 127.0.0.7 to 127.0.0.4. Later in the same capture, a PFCF Session Modification Request and corresponding Response are also visible between the same two addresses, showing that the SMF updated the session rules after the initial establishment.

Expanding one of the captured frames confirmed that the exchange took place over UDP, with both source and destination ports set to 8805, and that the Packet Forwarding Control Protocol layer was correctly identified by Wireshark's dissector — directly verifying that 127.0.0.4 and 127.0.0.7 correspond to the SMF and UPF respectively, communicating over the N4 interface realized on the loopback network in this single-host deployment.

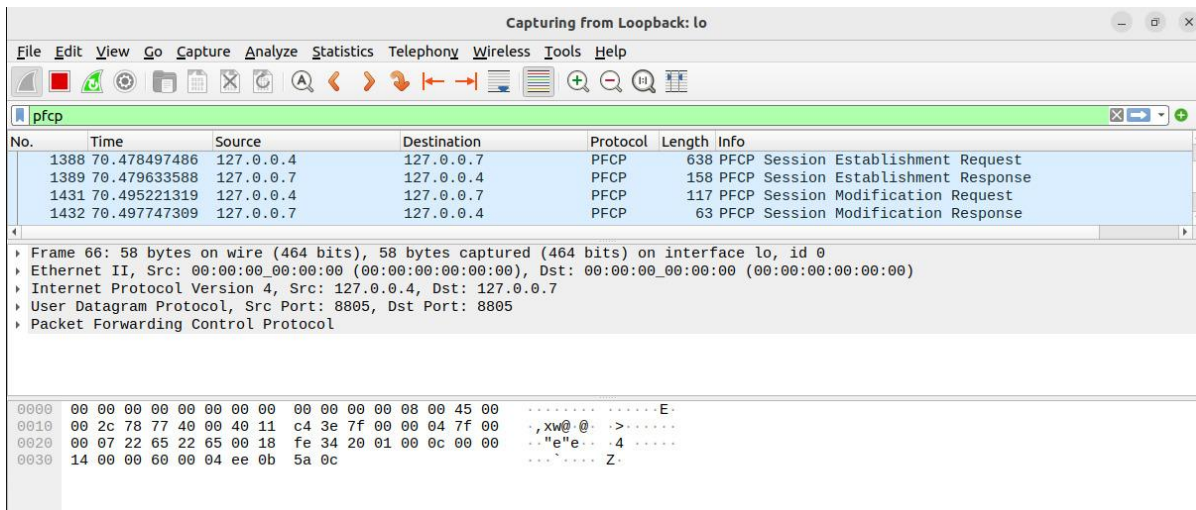


Figure 6 - Wireshark PFCP Packet Capture

9. Observation

- The subscriber with IMSI 999700000000001 was configured successfully in the Open5GS WebUI and showed an active (SERVICE_GRANTED) status.

- The UERANSIM gNB established an SCTP connection to the AMF and completed the NG Setup procedure successfully.
- The UE completed Initial Registration, including authentication and security mode configuration, and reached the MM-REGISTERED/NORMAL-SERVICE state.
- The PDU Session Establishment Request sent by the UE was accepted, and a TUN interface (uesimtun0) was brought up with IP address 10.45.0.3.
- Wireshark, filtered on pfcf over the loopback interface, captured the PFCP Session Establishment Request and Response exchanged between 127.0.0.4 (SMF) and 127.0.0.7 (UPF).
- A PFCP Session Modification Request and Response between the same two endpoints were also observed, indicating that the SMF updated session rules after establishment.
- All PFCP messages were confirmed to use UDP port 8805 on both ends, matching the standard PFCP port assignment.
- The control-plane signaling (registration, authentication, NAS/NGAP messages) and the user-plane session configuration (PFCP over N4) were observed as two distinct and independent message flows, consistent with the CUPS principle.

10. Result

The Control and User Plane Separation (CUPS) architecture of the 5G Core was successfully demonstrated. The PFCP Session Establishment Request and Response exchanged between the SMF and the UPF were captured using Wireshark, confirming that the N4 interface correctly carried the signaling required to configure the UPF for the UE's PDU session. The experiment verified that control-plane processing (SMF) and user-plane processing (UPF) operate as independent functions that communicate strictly through PFCP over N4.

11. Conclusion

This experiment successfully demonstrated the Control and User Plane Separation (CUPS) architecture of the 5G Core by building an Open5GS-based network with UERANSIM, completing UE registration and PDU session establishment, and using Wireshark to capture the PFCP Session Establishment Request and Response exchanged between the SMF and the UPF over UDP port 8805 on the N4 interface. The results confirmed that the control plane and user plane operate as independent functions coordinated entirely through PFCP, reinforcing the theoretical understanding of the SMF, the UPF, and the N4 interface through direct, hands-on observation.